

CHONGFENG LING

✉ career.ling@outlook.com

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Summary

AI Infrastructure Engineer specializing in LLM inference acceleration, CUDA kernel development, and quantized operator optimization on NVIDIA and Huawei Ascend platforms. Skilled in low-level performance tuning (GEMM, Attention) and integrating operator-level optimizations with runtime scheduling to enhance LLM serving performance.

Experience

FESCO Adecco Co., Ltd.

Nov. 2024 – Oct. 2025

AI Kernel Engineer (Ascend Platform)

Shanghai, China

Quantized MatMul Kernel: Ascend-based quantized MatMul kernel development.

- **Low-Level Operator Engineering:** Developed the *QuantBatchMatmul* kernel for Huawei Ascend A2/A3 chips. Leveraged heterogeneous hardware pipelines (Cube / Vector cores) to implement high-performance matrix multiplication through explicit multi-level memory management and asynchronous DMA scheduling.
- **Asymmetric Mixed Quantization:** Led the engineering deployment of **A8W8** / **A8W4** quantization for low-latency LLM inference. Designed an Adaptive Sliding Window Tiling strategy to resolve cache-misalignment bottlenecks under large *groupSize* workloads and improve hardware utilization.
- **Pipeline Overlapping & Buffer Optimization:** Reduced memory stall cycles by overlapping AIC/AIV execution through **ping-pong buffering** and asynchronous DMA scheduling, mitigating Scale-memory bottlenecks.
- **Compute Overhead & Accuracy Optimization:** Optimized **Offset Compensation** in asymmetric quantization by mapping dense MAC operations onto AIV *ReduceSum/Broadcast* vector instructions, while tightly controlling dequantization rounding to achieve **100% pass rates** in LLM accuracy validation.

Mathartsys, Inc.

Feb. 2021 – Sep. 2021

NLP Engineer Intern

Shanghai, China

Semantic Analysis System: Automotive supply chain semantic analysis system development.

- Developed a BERT/Transformer-based semantic analysis system for automotive supply chain text, enabling automated text classification, information extraction, and semantic understanding of unstructured documents.

Projects

CUDA-Handson | *CUDA, Nsight Compute, Nsight Systems* [Github]

- **High-Performance GEMM Optimization:** Achieved **80%+ cuBLAS FP32 throughput** on RTX 5080-class GPUs through shared-memory tiling, register blocking, and vectorized memory access.
- **Attention Kernel Optimization:** Implemented and optimized Flash Attention with shared-memory tiling and single-pass online softmax, eliminating the need for full $N \times N$ matrix materialization. Integrated the custom kernel into the nano-vLLM backend, ensuring zero end-to-end performance degradation.
- **Performance Profiling & Tuning:** Conducted kernel-level bottleneck analysis using **Nsight Compute** and the **Roofline Model**, balancing register pressure against **SM occupancy**.

Memory-Aware GEMM & SpMV Engine | *CUDA, Numba, Python* [Github]

- **GEMM Optimization:** Designed parallel Numba CUDA kernels for PDE workloads. Leveraged contiguous memory layouts to improve SIMD vectorization efficiency and cache coalescing behavior.
- **SpMV Engine & Profiling:** Implemented a JIT-compiled SpMV engine with a custom CSR format. Combined parallel COO assembly with FEM profiling to analyze MINRES convergence bounds.

High-Performance N-Body Simulation | *C++, OpenMP, MPI* [Github]

- **N-Body Simulation Engine:** Built a high-performance N-body planetary simulator in C++, achieving **50x acceleration** through memory-layout optimization and compiler-level tuning while benchmarking energy loss across varying timestep sizes.
- **Parallel Performance Analysis:** Implemented multi-core parallelism using OpenMP and conducted comprehensive **strong** and **weak scaling** analyses to identify synchronization bottlenecks and thread-creation overheads.

Education

University College London

Sep. 2022 – Dec. 2023

MSc in Scientific and Data Intensive Computing

London, UK

University of Liverpool

Jun. 2022

BSc in Applied Mathematics

Liverpool, UK

Xi'an Jiaotong-Liverpool University

Sep. 2017 – Jun. 2022

BSc in Applied Mathematics

Suzhou, China

Skills

Languages : C++ (CUDA, Ascend C), Python

Frameworks/Tools : PyTorch, CUTLASS, vLLM, Nsight Systems (nsys), Nsight Compute (ncu), Linux, Git

Last updated on 2026-05